



Society of Petroleum Engineers

SPE-229068-MS

Experimental Determination of Triethylene Glycol (TEG) and Methanol Loss in Compressed CO₂

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229068-MS](https://doi.org/10.2118/229068-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

The presence of polar organic impurities in dense-phase CO₂ streams is a critical issue for carbon capture, utilization, and storage (CCUS) and enhanced oil recovery (EOR) operations. Among these impurities, triethylene glycol (TEG) and methanol are particularly important, owing to their widespread use in gas dehydration and hydrate inhibition. Their persistence in CO₂ can alter compression behavior, promote corrosion through water film formation, and contaminate storage reservoirs.

This study investigates the phase behavior of TEG and methanol in compressed CO₂ under pipeline-relevant conditions using a high-pressure gas–liquid saturation unit combined with nuclear magnetic resonance (NMR) spectroscopy. The methodology enables reproducible equilibration and precise detection of trace polar organics in the vapor phase. The observed solubility trends are interpreted in terms of operational risks, including condensation, foaming, and corrosion initiation.

The work does not disclose numerical data but provides a technical framework for understanding how methanol and TEG behave in dense-phase CO₂ systems, linking laboratory observations to practical challenges in transport and storage. The findings are directly relevant for defining impurity specifications, improving EoS parameterization, and informing risk-based design and monitoring strategies in CCUS projects.

Introduction

As carbon capture, utilization and storage (CCUS) infrastructure expands from pilot-scale demonstrations into complex, multi-node industrial networks, ensuring reliable CO₂ transport is becoming a critical design and operational challenge. In emerging industrial clusters, capture plants from diverse sectors such as power generation, cement production, hydrogen manufacture, and steelmaking are expected to feed into shared CO₂ pipelines. These pipelines extend hundreds of kilometers and operate in the dense phase to maximize transport efficiency (Patchigolla & Oakey, 2013) (de Visser, et al., 2008). Unlike isolated projects, these multi-user networks must handle streams from different capture technologies with varying impurity profiles, complicating specification settings and operational control. Although dense-phase transport is preferred

for efficiency, at high pressures even trace amounts of contaminants can shift phase boundaries, promote corrosion, or compromise material compatibility. Maintaining high CO₂ purity is therefore essential not only to ensure storage integrity but also to safeguard compression systems, prevent foaming or hydrate formation, and protect the long-term reliability of pipeline infrastructure. In addition, shared networks create commercial challenges, as operators must enforce entry specifications, allocate responsibility for off-spec CO₂, and implement monitoring systems capable of detecting and mitigating any deviations in real time.

Impurities in CO₂ streams arise from capture, conditioning, and transport processes, and their type and concentration depend strongly on the feed gas composition and the separation technology employed. (Porter, Fairweather, Pourkashanian, & Woolley, 2014) Common examples include water vapor, oxygen, nitrogen, sulfur compounds such as SO₂ and H₂S, and trace hydrocarbons or process-derived organics. While some impurities are carried over from the feedstock or flue gas, others originate from solvents, inhibitors, or processing chemicals used during capture and handling. Their presence can influence almost every aspect of CCS performance, from corrosion and hydrate formation in pipelines to injectivity and long-term storage integrity in reservoirs. Industrial specifications therefore set limits for broad impurity classes, including non-condensable gases (e.g., N₂, O₂), acid gases (SO₂, NO_x, H₂S), and water, with a residual category of "other components" typically covering organics and particulates.

Within this latter category, the behavior of trace polar organics in dense-phase CO₂ remains less understood. These compounds are not rare contaminants. They arise directly from standard process chemicals used in CO₂ capture and handling. Triethylene glycol (TEG) is the most common glycol employed in dehydration units due to its high boiling point, low volatility, and strong water absorption capacity. Methanol, on the other hand, is favored as a hydrate inhibitor in subsea production and cold-climate operations due to its low cost, high effectiveness at small dosages, and full miscibility with water. Both TEG and methanol are polar compounds capable of interacting with trace water and other impurities. Even at parts-per-million (ppm) levels, they can influence gas phase behavior, hydrate formation, raise the water dew point, and contribute to corrosion under wet conditions. (Ahmadi, Chapoy, Fahmi, & Teberikler, 2022) (Chapoy, et al., 2013) (Wang, Ryan, Anthony, Wildgust, & Aiken, 2011) Their solubility and persistence in dense-phase CO₂ are strongly governed by the pressure and temperature profiles typical of CCS pipelines. As a result, small upstream introductions of these chemicals can persist over long distances, potentially affecting downstream equipment not designed for such exposure.

The mechanisms by which these organics enter CO₂ streams are well recognized. In dehydration units, fine droplets or vapor-phase TEG may pass into the treated gas when separator or mist eliminator performance is compromised, while methanol frequently enters through hydrate control operations, especially during transients when recovery is incomplete. Once present, both compounds are persistent: methanol redistributes dynamically with pressure and temperature, whereas TEG tends to remain as stable films or dissolved at trace levels. Their presence is not usually critical by itself, but in combination with water and acid gases they can significantly raise the water dew point, destabilize separation processes, and accelerate corrosion. These risks are particularly relevant under modern CCS pipeline conditions, which typically operate above 70 bar to maintain CO₂ in the dense phase, with temperature profiles dictated by ambient climate, terrain, and burial depth (Patchigolla & Oakey, 2013). Under such conditions, the high density of the CO₂ stream enhances the dissolution of polar organics, altering their behavior compared to natural gas systems and making them more challenging to predict and control.

These behaviors have direct implications for CCS transport operations. Solubility data for methanol and TEG in CO₂ enable more accurate prediction of dew-point shifts, foaming risks, and film formation under different operating pressures and temperatures. This information supports specification setting at pipeline entry points, informs separator and compressor design, and underpins monitoring programmes that target ppm-level impurities. By linking laboratory solubility measurements with industrial operating scenarios, the work presented here provides a foundation for risk-based impurity management in CCS pipeline systems.

Experimental Methods

Solubility measurements were performed using a state-of-the-art configuration that comprises a dynamic gas–liquid equilibration system designed to replicate CCS pipeline conditions and an advanced analytical approach. The gas–liquid equilibration system is presented in Fig. 1.

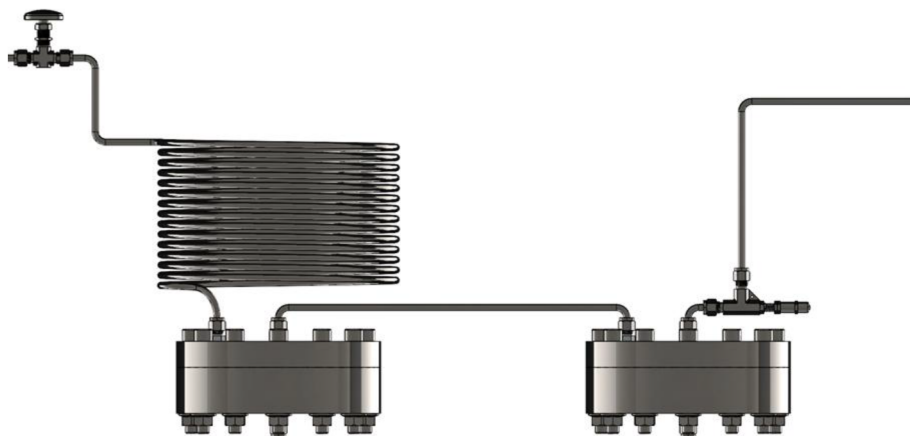


Figure 1—High Pressure and High Temperature Gas Saturating Unit Used for the Study

The experimental procedure started by injecting a certain volume of organic compound (TEG or methanol) into the bottom side of saturating unit (liquid bath). Then, the system was pressurized to target conditions using CO₂. The system consists of a two-stage saturation unit, the role of which is to ensure sufficient time for gas–liquid contact and subsequent equilibration. Once the system is pressurized, the flow is initiated. Prior to entering the saturation unit, gas stream is pre-heated through the coiled heat exchanger. The saturated gas then exits the system. The outlet is equipped with a high-pressure liquid trap to ensure no excess fluid was picked up by CO₂ stream.

The single-phase vapor is then flashed through a gas scrubber equipped with a porous plate on the tube to facilitate an even distribution of gas bubbles raising up towards the top. The scrubber solvent selected for CO₂ purification is deuterated dimethyl sulfoxide (DMSO-*d*₆). Several quality checks proved that the scrubbing efficiency of this setup is 100%. Once the experiment is performed and all the TEG/Methanol scrubbed from the saturated CO₂, a subsample is taken and placed into NMR tubes. High frequency (600 MHz) NMR spectroscopy was used to quantify the captured amounts of either TEG or methanol.

An internal standard, durene, allowed accurate concentration determination without external calibration. The combined expanded uncertainty for TEG and MeOH quantification was estimated at 6.4% (95% confidence level), accounting for fluid purities, mass flow accuracy, solvent volume measurement, and NMR integration precision.

The experiments were performed at temperatures from 25 °C to 90 °C and pressures from 1 bar to ~110 bar. The mass of dry CO₂ was measured using high precision mass flow meter operating based on the Coriolis flow measurement principle. Once the amount of captured impurity was quantified, vapor phase composition of saturated CO₂ was then calculated.

At this stage, these trials were not intended to produce a full thermodynamic dataset but to identify operationally significant behaviors. Full apparatus schematics, calibration records, and datasets will be reported in a forthcoming journal article. Here, the emphasis is on the behavioral patterns most relevant to CCS transport.

Results and Discussion

Experimental data and benchmarking against literature data is presented in Fig. 2 for TEG in CO₂ and in Fig. 3 for methanol in CO₂.

The solubility values obtained in this work show relatively good agreement with the literature data reported by (Yonemoto, Charoensombut-Amon, & Kobayashi, 1990), particularly at low to intermediate pressures where both datasets consistently indicate concentrations below 10 ppm. At higher pressures, the sharp increase in solubility observed in literature is also reproduced in the present measurements, with only minor deviations in magnitude. These small differences may reflect variations in experimental configuration. The present work utilized a staged saturation approach with effective droplet trapping to minimize any artefacts, mostly involving droplet entrainment before reaching the analytical tool. The close correspondence between the two datasets provides confidence in the reliability of the current measurements and confirms that TEG remains only sparingly soluble in CO₂ at 40 and 60 °C, with concentrations in the ppm range across the pressure conditions relevant to pipeline transport.

The solubility of methanol in CO₂ at 25 °C measured in this work is also in good agreement with literature values reported by Katayama and Ohgaki, and Brunner and co-workers across the investigated pressure range. (Katayama & Ohgaki, 1976), (Brunner, Hültenschmidt, & Schlichthärle, 1987) Both datasets reproduce the characteristic U-shaped pressure dependence, with higher solubility at low pressures followed by a decrease as pressure increases toward the dense-phase region. The present measurements overlap closely with the literature between 20 and 60 bar, confirming the reproducibility of this trend and the reliability of the applied NMR quantification method. Minor deviations at very low pressures (<15 bar) may arise from differences in equilibration time or sampling efficiency, as methanol's volatility makes accurate quantification more sensitive under these conditions.

Overall, the consistency between the present measurements and literature data for both TEG and methanol reinforces the validity of the experimental approach and highlights the reliability of the results across two chemically distinct polar organics. For methanol, the agreement confirms the robustness of solubility data under near-ambient conditions, while for TEG, the close correspondence with prior studies demonstrates that the applied methodology effectively minimizes artefacts such as droplet entrainment. Taken together, these findings support the use of the present dataset as a benchmark for evaluating thermodynamic models and provide a sound basis for improving impurity management strategies in CCS applications.

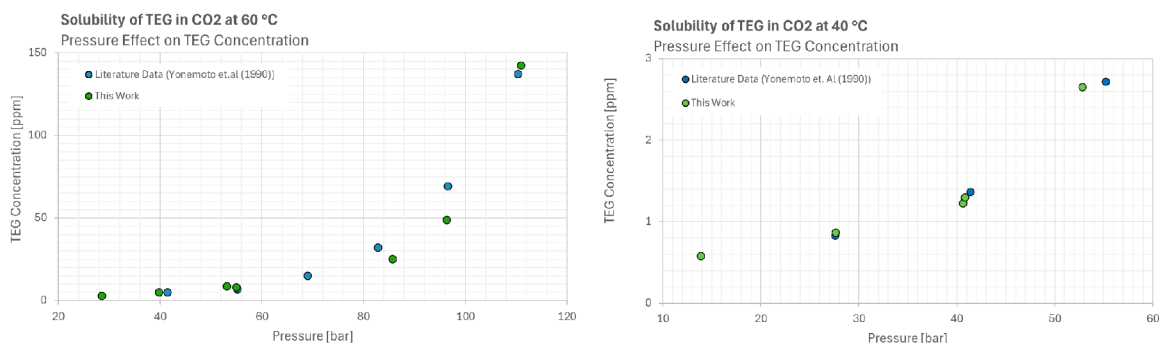


Figure 2—a (left), b (right) – (a) Comparison of experimental data within this work and literature data obtained from Yonemoto et.al (1990) at 60°C. (b) Comparison of experimental data within this work and literature data obtained from Yonemoto et.al (1990) at 40°C.

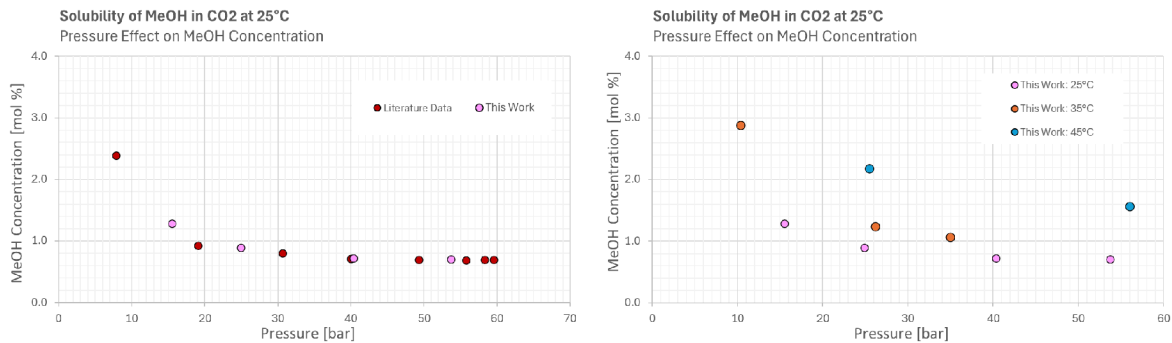


Figure 3—a (left), b (right) – (a) Comparison of experimental data within this work and literature data obtained from Katayama and Ohgaki (1976) at 25°C. (b) Summary of experimental data of methanol solubility in CO2 at 25°C, 35°C and 45°C.

Methanol displayed a non-linear solubility trend with pressure, with a distinct minimum around 40–60 bar. This means that small pressure changes in this range can significantly alter vapor-phase methanol concentrations. For example, a maintenance-related depressurization could inadvertently shift the system into a higher-solubility range, increasing methanol content downstream and potentially disrupting separation or hydrate control. By contrast, TEG solubility increased steadily with both pressure and temperature. While absolute vapor-phase concentrations were lower than for methanol, they were sufficient to risk downstream deposition and fouling. This is particularly relevant in cold sections, such as buried winter pipelines or subsea lines exposed to seasonal seabed cooling. TEG-rich films can trap water and acid gases, fostering corrosion.

The differences between these compounds are summarized in Tab. 1, which provides a quick reference for volatility, phase behavior, deposition risk, corrosion mechanisms, and removal difficulty. This comparison highlights the need for tailored mitigation and monitoring strategies.

Table 1—Key reservoir parameters of individual layers

Property/Behaviour	Methanol (MeOH)	Triethylene Glycol (TEG)
Volatility	High	Low
Pressure effect	Non-linear; solubility minimum	Monotonic increase
Temperature effect	Moderate increase	Strong increase
Condensation risk	Mainly during rapid cooling	Persistent cold-film formation
Corrosion risk mechanism	Alters water chemistry	Creates microenvironments
Removal difficulty	Moderate	High once dissolved

Operational Implications in CCS

The presence of methanol and TEG in dense-phase CO2 pipelines has direct consequences for design, operation, and monitoring. Managing these risks requires a balance of prevention, mitigation, and early detection.

Prevention

The first line of defence is to minimise the entry of methanol and TEG into CO2 pipelines and to avoid operating conditions that amplify their impact. From a design perspective, thermal and hydraulic

simulations should explicitly account for impurity effects on phase behaviour, ensuring that compressor staging, separator sizing, and material selection reflect the risks of dew-point shifts and solubility changes. Reliable separation technology is particularly important for reducing TEG carryover, with well-designed mist eliminators and regular maintenance preventing entrainment under both steady and transient flow conditions. Hydrate control programmes should include planned recovery steps to limit residual methanol, while operating procedures should avoid prolonged residence in the pressure ranges where methanol exhibits a solubility minimum. Together, these measures aim to keep impurity levels as low as possible and to reduce the likelihood of unfavourable phase redistributions during normal operation.

Mitigation

When methanol and TEG are already present, the focus shifts to limiting their damaging effects on equipment and infrastructure. Corrosion management must reflect the distinct behaviour of each compound. Methanol, due to its volatility, can redistribute during pressure changes and weaken inhibitor films, requiring inhibitor systems that are specifically formulated and regularly monitored for stability in methanol-rich environments. TEG, on the other hand, promotes cold-film formation and microenvironmental corrosion, which makes downstream protection measures essential. In vulnerable sections such as cold subsea lines or buried pipelines, corrosion-resistant alloys or advanced protective coatings provide longer-term integrity than carbon steel alone. Localised treatments, including purge or warm-up cycles, can also help reduce the persistence of TEG films in systems where contamination cannot be fully prevented. These targeted mitigation strategies ensure that even when impurities reach the network, their adverse impacts are contained.

Early Detection

Continuous monitoring of impurity levels and inhibitor performance is essential to ensure these mitigation strategies remain effective. Sampling points should be placed strategically along the network, not only at pipeline entry points but also downstream of compression and cooling stages where phase changes are most likely. Fast analysers capable of detecting ppm-level organics in dense-phase CO₂ could provide rapid feedback for fast decision making. In multi-user pipelines, coordinated monitoring programmes are critical for identifying contamination sources and avoiding disputes between operators. Such programmes should include standardised reporting protocols, agreed alarm thresholds, and transparent data sharing to support rapid decision-making. In addition, integrating impurity monitoring with corrosion probes, weight-loss coupons, or electrochemical sensors can provide early warning of localised attack before significant damage occurs. Linking these datasets with operational logs, such as hydrate control injections, separator maintenance, or pressure cycling, helps operators distinguish between normal fluctuations and events that require intervention.

Looking ahead, the effectiveness of monitoring programmes can be greatly enhanced by incorporating big data processing and predictive analytics. (Fang, Cheng, Gai, Lin, & Lou, 2023) By compiling long-term datasets from impurity measurements, corrosion sensors, and operational logs, machine-learning algorithms can be trained to identify subtle patterns that precede equipment upsets or integrity failures. Such tools allow operators not only to react to off-spec events but also to forecast when and where problems are most likely to occur, enabling proactive maintenance and optimisation of inhibitor use. Over time, these data-driven approaches can transform impurity management from a reactive process into a predictive framework, improving reliability and reducing costs across CCS transport networks.

Summary

Methanol and TEG, even at trace levels, pose distinct operational challenges in dense-phase CO₂ transport. Managing these risks requires a structured approach. Prevention begins with design measures that account for impurity-driven phase behavior and incorporate robust separation technologies and suitable materials

in high-risk zones. Mitigation focuses on reducing TEG carryover, selecting inhibitors compatible with methanol, and applying corrosion-resistant alloys or advanced coatings where exposure is unavoidable. Early detection relies on strategically placed sampling points, fast-response impurity analyzers, and coordinated monitoring programs across multi-user networks.

Together, these measures form a practical framework for controlling impurity risks in CCS transport. By implementing prevention, mitigation, and monitoring in an integrated manner, operators can maintain system reliability and reduce the likelihood of unplanned downtime or asset degradation.

Conclusions

This study has provided new experimental insight into the solubility of methanol and triethylene glycol in dense-phase CO₂ under conditions representative of CCS transport. The results demonstrate that methanol exhibits a non-linear solubility trend with a minimum in the intermediate pressure range, whereas TEG shows a monotonic increase in solubility with both pressure and temperature, albeit at lower absolute concentrations. Despite these differences, both compounds display behaviours that can complicate pipeline operation by raising the water dew point, promoting localised corrosion, or destabilising separation and compression processes.

The close agreement with available literature confirms the robustness of the applied methodology, which combines a high-pressure saturation system with quantitative NMR analysis, while minimising artefacts such as droplet entrainment. This validation supports the use of the present dataset as a benchmark for improving thermodynamic models and for refining specifications that govern impurity control in multi-user CO₂ networks.

Beyond the laboratory findings, the work highlights the broader operational implications of polar organics in CCS transport. Integrating accurate solubility data into design tools, operational procedures, and monitoring strategies enables operators to better anticipate impurity-driven risks and to develop more reliable corrosion management frameworks. In the longer term, coupling such datasets with digital monitoring and predictive analytics will help transform impurity management from a reactive process into a proactive, risk-based approach.

By bridging fundamental measurements with practical guidance, this work contributes both to the scientific understanding of CO₂–organic interactions and to the operational knowledge base required for safe and efficient deployment of CCS infrastructure at scale.

Acknowledgements

This work was carried out as part of the Pace CCS collaborative research program. The authors thank Pace CCS for project funding and strategic guidance, and fluidXlab GmbH for access to state-of-the-art high-pressure facilities and invaluable technical support during the experimental campaign.

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